①

② matrix trans. → linear trans.

③ . . . → inner operators

④ dot product → inner product $V \times V \to \mathbb{C}$

⑤ symmetric matrix $A^T = A$ $A \in M_{n\times n}(\mathbb{R})$

→ self-adjoint operators $T: V \to V$

$T^* = T$

generize $\mathbb{R}^n \to V$

① Laplacian operator $\Delta: V \to V$ (mean?)

$$\Delta(f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y} + \frac{\partial^2 f}{\partial z}$$

(f: func.)

② Laplace equation: $\Delta(f) = 0 = 0(f)$

$\Leftrightarrow 0$: the eigenspace of Δ

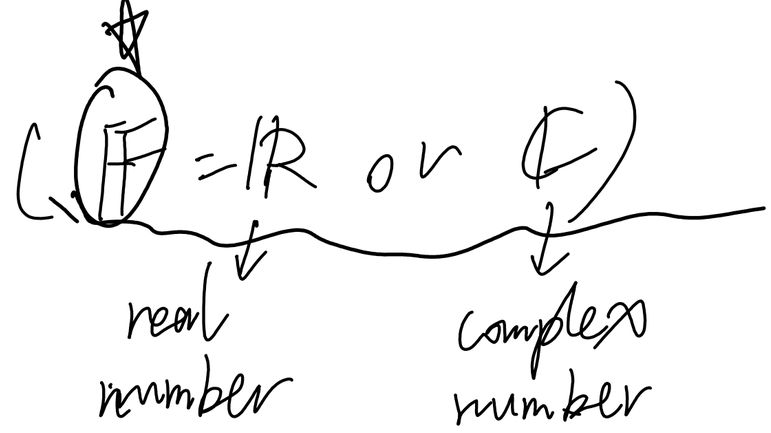
② Schrödiglr operator $\hat{H}: V \to V$

~~④~~ $\hat{H}(f) = \overline{\quad - - \quad}$

量子力学中: $\hat{H}(f) = \lambda f$

Vector Space

Def.: Let $\mathbb{F}$ be a field ($\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$)



A <u>vector space</u> <u>over $\mathbb{F}$</u> is
a set V equipped with:
① scalar multiplication $\mathbb{F} \times V \to V$
$(\alpha, \underline{v}) \mapsto \alpha \cdot \underline{v}$
② vector addition $V \times V \to V$
$(\underline{v}, \underline{w}) \mapsto v + w$
s.t. $\cdot$ and $+$ satisfy the <u>8 axioms</u> (見于2040)

e.g. ① $V = \mathbb{R}^n$
is a vector space over $\mathbb{F} = \mathbb{R}$

e.g. $\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} + \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = \begin{pmatrix} a_1+b_1 \\ \vdots \\ a_n+b_n \end{pmatrix}$; $\alpha \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} \alpha a_1 \\ \vdots \\ \alpha a_n \end{pmatrix}$

$\underline{0} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$; $V = \mathbb{C}^n \cdots \mathbb{R}$

② $V = M_{m \times n}(\mathbb{F})$ is a v. space over $\mathbb{F}$)

e.g. $\alpha \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} = \begin{pmatrix} \alpha a_{11} & \cdots \\ \vdots & \end{pmatrix}$

$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} + \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{m1} & \cdots & b_{mn} \end{pmatrix} = \begin{pmatrix} a_{11}+b_{11} & \cdots \\ \vdots & \\ (& \end{pmatrix}$

③ $V = C^\infty(\mathbb{R}^n)$ is a v. space over $\mathbb{R}$

(or $\mathbb{R}$): $C_c(\mathbb{R}^n)$,

all cont. func. $f: \mathbb{R}^n \to \mathbb{R}$

--- (未)

④ $V = P(\mathbb{R})$

(or $\mathbb{R}$): $\mathbb{R}[x]$,

all polynomials with real coefficients

of degree $\leq n$

(deg([polynomial]) = degree of)

--- (未)

zero vector: $\underline{0} = 0 + 0x + \cdots + 0x^n$

Vector Subspace

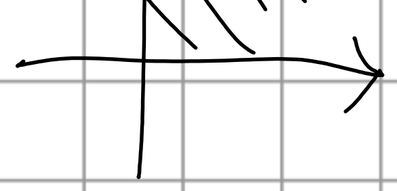
Def.: Let V be a vector space over $\mathbb{F}$

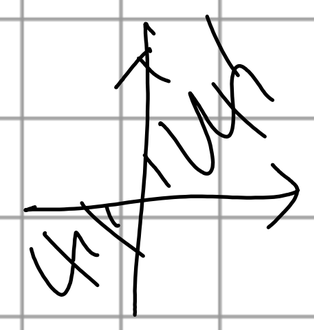
a vector subspace of V

is a subset $W \subseteq V$

if W is

a vector space using the operations $\cdot$ and $+$ in V

e.g. ① $V = \mathbb{R}^2$, $W =$ 

② $V = \mathbb{R}^2$, $W_2 =$ 

$(-3) \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -3 \\ -6 \end{pmatrix}$, but $\begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} -1 \\ -5 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$

$\in W_2$ $\in W_2$ $\notin W_2$

W_2 is not a V. subspace of V

(From 2040)

all subspaces $W \leq \mathbb{R}^2$ are of the form $\mathrm{Span}_{\mathbb{R}}\{\underline{v}_1, \cdots, \underline{v}_k\}$

for some $\underline{v}_1, \cdots, \underline{v}_k$ in $\mathbb{R}^n$

(?)

(define $\mathrm{Span}(S)$ for $\forall S \subseteq V$, $\forall V$ next lecture

(Theorem)

Let V be a vector space

A subset $W \subseteq V$ is a subspace of V

if: ①

(and) |

②

$\forall w_1, w_2 \in W,\ \alpha, \beta \in \mathbb{R},\ \alpha w_1 + \beta w_2 \in W$